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Chimps, Humans 96 Percent the Same, Gene Study Finds

Stefan Lovgren
for [National Geographic News](#)
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Scientists have sequenced the genome of the chimpanzee and found that humans are 96 percent similar to the great ape species.

"Darwin wasn't just provocative in saying that we descend from the apes—he didn't go far enough," said Frans de Waal, a primate scientist at Emory University in Atlanta, Georgia. "We are apes in every way, from our long arms and tailless bodies to our habits and temperament."

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Because chimpanzees are our closest living relatives, the chimp genome is the most useful key to understanding human biology and evolution, next to

the human genome itself. The breakthrough will aid scientists in their mission to learn what sets us apart from other animals.

By comparing human and chimpanzee genomes, the researchers have identified several sequences of genetic code that differ between human and chimp. These sequences may hold the most promise for determining what creates human-specific traits such as speech.

"If people are asking what makes us human, they're not going to find a smoking gun [in this study]," said Evan Eichler, a genome scientist at the University of Washington in Seattle who was part of the research team. "But they're going to find suggestions for where to look."

The project was conducted by an international group of scientists called the Chimp Sequencing and Analysis Consortium. Sixty-seven researchers co-authored the study, which is detailed tomorrow in the journal *Nature*.

Genetic Blueprints

To map the chimp genome, researchers used DNA from the blood of a male common chimpanzee (*Pan troglodytes*) named Clint, who lived at the Yerkes National Primate Research Center in Atlanta. Clint died last year from heart failure at the relatively young age of 24.

A comparison of Clint's genetic blueprints with that of the human genome shows that our closest living relatives share 96 percent of our DNA. The number of genetic differences between humans and chimps is ten times smaller than that between mice and rats.

Scientists also discovered that some classes of genes are changing unusually quickly in both humans and chimpanzees, as compared with other mammals. These classes include genes involved in the perception of sound, transmission of nerve signals, and the production of sperm.

Despite the similarities in human and chimp genomes, the scientists identified some 40 million differences among the three billion DNA molecules, or nucleotides, in each genome.

The vast majority of those differences are not biologically significant, but researchers were able to identify a couple thousand differences that are potentially important to the evolution of the human lineage.

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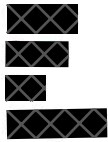


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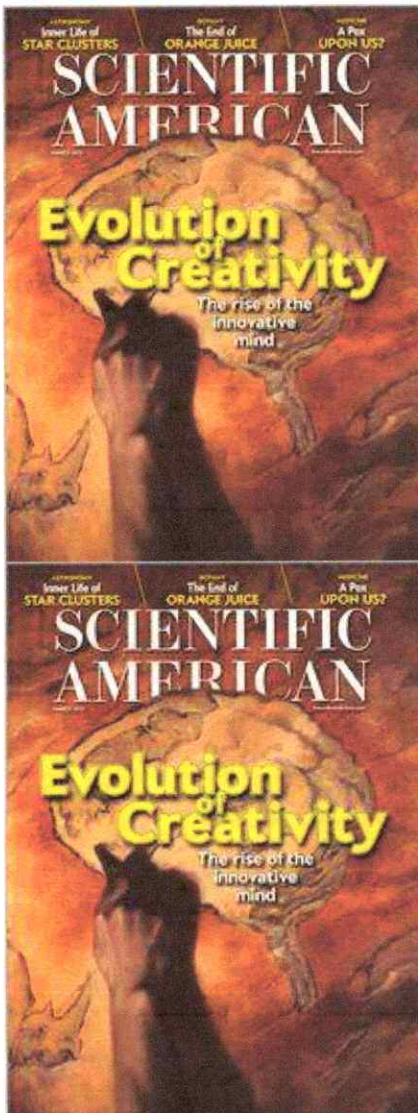
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What does the fact that we share 95 percent of our genes with the chimpanzee mean? And how was this number derived?

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Prescott Deininger of the Tulane [Cancer](#) Center in New Orleans explains.

There is a significant body of evidence that supports the idea that the chimpanzee is the closest genetic relative of humans. This was first determined through a large number of studies, some of which used genomic DNA hybridization to detect the level of sequence mismatches, as well as analyses of individual protein molecules. These early findings suggested that chimps and humans might typically have sequences that diverge from one another by only about 1 percent.

We now have large regions of the chimpanzee genome fully sequenced and can compare them to human sequences. Most studies indicate that when genomic regions are compared between chimpanzees and humans, they share about 98.5 percent sequence identity. The actual relationship depends on what types of sequences are being compared and the size of the comparison unit. A report published in the *Proceedings of the National Academy of Sciences* in 2002 suggested that under the most rigorous alignments, the match would be only 95 percent similarity overall. This resulted from the researchers treating changes involving small insertions and deletions of bases differently than previous investigators did over a very large region. A few questions still remain as to whether the

chimpanzee genome sequence data are of high enough quality at this point for reliable comparison. In general, however, the overall conclusion is that most genes would share about 98.5 percent similarity. The actual protein sequences encoded by these genes would then typically be slightly more similar to one another, because many of the mutations in the DNA are "silent" and are not reflected in the protein sequence.

Given the very strong similarity between the chimpanzee and human genomes, many people wonder how we can be so different. At this point, there have been only a few isolated examples of genes that are functionally present in chimpanzees but not in humans, and vice versa. Thus, chimps and humans may share as many as 99.9 percent of the same genes with most of those genes being 99 percent similar in their sequences. Chromosomes do not exhibit big structural differences either. Although there are a number of small chromosomal changes that rearrange the order of genes on regions of those chromosomes, most of these are thought to leave gene function unchanged. It seems likely that the differences between human and chimpanzee phenotypes depend more on subtle regulatory changes more than on the presence of different genes. For instance, it may be that there are changes in some genes that alter the amount of protein produced by that gene at different stages in the development of a chimp versus a human. Alternatively, there may be small changes to the structures of the proteins (from the 1 percent divergence) that produce changes in how they interact with other cellular components and therefore subtly alter the pathways in which they are involved. At this point, we do not know which types of changes are responsible for the relatively big differences between chimps and humans.

It is worth noting that individual humans generally differ by about 0.1 percent genetically. Thus, chimps differ from humans by about 15-fold more, on the average, than humans do from one another. The 0.1 percent human divergence certainly results in significant variation in physical appearance and traits between different humans. Therefore, perhaps we shouldn't be so surprised that chimps could be 98.5 percent related to humans. Relatively small genetic changes can produce major phenotypic changes.

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